

Mr. Harendra Kumar Dadhich is a seasoned professional with a B. Tech. in Electronics and Communication Engineering (ECE) and over 12+ years of experience in the Geospatial domain, including 7+ years in **Data Analytics, Data Science, and Predictive Modelling**. His expertise spans the development and analysis of Machine Learning models such as Regression, Classification, and Clustering, as well as Deep Learning models.

Mr. Dadhich is proficient in operating systems like Linux and Windows, and skilled in programming languages such as Python and Shell Scripting. As a keen planner and strategist, he excels in ensuring smooth operations and enhancing efficiency through cutting-edge technologies and business reviews. His portfolio includes successful projects in Satellite Meteorology, GPS Technology for crustal deformation mapping, fraud transaction detection, network intrusion detection systems, and **AI-ML-driven projects**. He worked over the Microwave Remote Sensing applications like **Natural Disaster (Cyclone) detection and tracking, Oil Spill detection and Tsunami Detection**. He is an achievement-oriented professional with strong decision-making and problem-solving skills, dedicated to delivering effective solutions.

Currently, Mr. Dadhich is working at **LTIMindtree** as a Senior Specialist in the **Geospatial** domain. He has developed **crop type classification machine learning models** using open-source remote sensing data across multiple cropping seasons and geospatial data. He demonstrated comprehensive machine learning skills, from data collection to deployment, with a strong understanding of agricultural systems. He leveraged advanced geospatial and coding expertise, including proficiency in **Python, Google Earth Engine, and GIS**, to create an effective crop type classification system using ML models.

His professional experience includes roles at **BirlaSoft**, where he developed predictive models, applied machine learning algorithms, and created comprehensive reports and dynamic dashboards. At the **National Centre for Seismology, Ministry of Earth Science, New Delhi**, he worked on crustal deformation mapping, earthquake intensity map generation, and prediction using ML and deep learning models, and mentored students. At the **International Center for Radio Science**, he applied advanced modelling techniques for oceanography, land, seismological, and atmospheric applications of microwave remote sensing.

He had extensive experience in analyzing ground truth data, implementing the Radiative Transfer Model, and developing models for applications such as natural disaster detection and tracking, oil spill detection, and tsunami detection. He worked on the project for soil moisture determination over India for the **Indian Meteorological Department (IMD)**, retrieval of salinity from satellite data for **Indian National Centre for Ocean Information Services (INCOIS)**. He utilized various satellite-based software tools. His work significantly contributed to the advancement of **remote sensing applications** and demonstrated his strong analytical and technical skills. He used remote sensing data for various applications of microwave remote sensing, enhancing his programming skills and computational methods.

Mr. Dadhich has completed several certifications, including Data Science and Machine Learning using Python from AnalytixLabs, Gurgaon, and ArcGIS: Introduction to GIS from ESRI India. He has also undertaken courses on **AWS Machine Learning**, Power BI, Calculus for **Machine Learning**, Machine Learning Regression, and Generative AI from platforms like Coursera and Google Cloud.

Mr. Dadhich has contributed to numerous publications, including “Overview of Artificial Intelligence (AI) and Machine Learning (ML) in Seismology” published in Geo Hazards: Analysis, Modelling, and Forecasting, and “Ground Motion Prediction Equation for NW Himalaya and its Surrounding Region” in **Quaternary Science Advances**. He has also co-authored papers on geoelectric characterization, ground-motion parameters, and oil spill detection using satellite data. His work has been published in reputable journals such as **Scientific Reports**, **Tectonophysics**, and the Indian Journal of Radio and Space Physics.

Additionally, he has held professional memberships and positions of responsibility, including **Joint Secretary** in the 9th International Conference on Microwaves Antenna Propagation & Remote Sensing (**ICMARS-2013**), **Treasurer of the Executive Council of IMS Jodhpur** Chapter, IEEE GRSS Member, **IMS Life Member**, and reviewer for **IGARSS** and **JGR International Journal**.